

SOC ALERT MONITORING & LOG CORRELATION

Name: Kadiyala Revanth Kumar

Agenda

- Project Overview
- Objectives
- SIEM & Log Source Research
- Lab Architecture
- Data Collection
- Alert Generation & Monitoring
- Dashboard Overview
- Log Correlation & Investigation Workflow
- AI-Powered Incident Response Enhancement
- Impact Analysis & Recommendations
- Conclusion
- References

Abstract and Objectives

- **Abstract**

- Developed a SOC monitoring environment using Splunk and Sysmon.
- Collected authentication, endpoint, and network security logs.
- Generated security alerts and performed log correlation.
- Investigated suspicious activities using SOC analyst workflows.

- **Objectives**

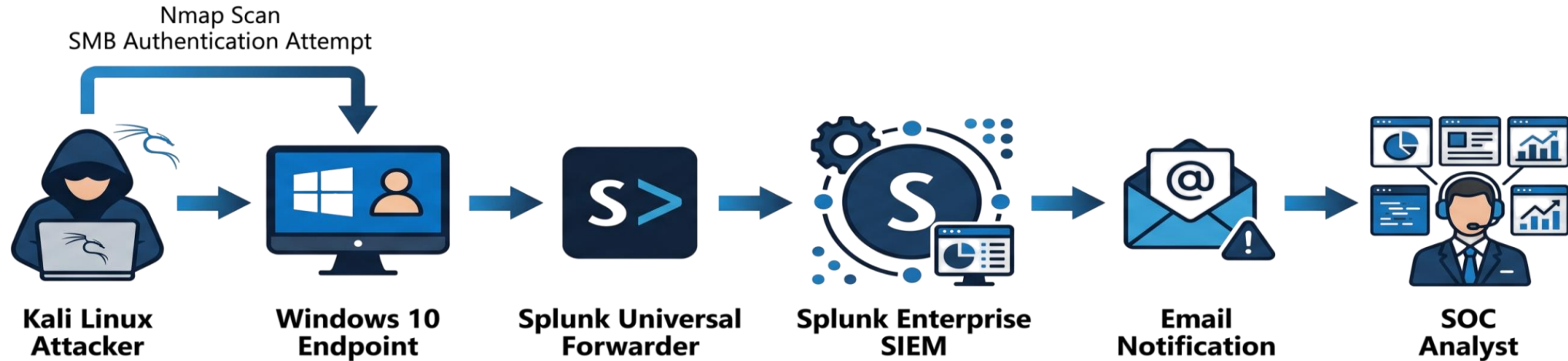
- Monitor security events in real time.
- Generate and investigate security alerts.
- Correlate multiple events to identify attack patterns.
- Improve visibility into suspicious system activities.

SIEM & Log Source Research

- **Splunk Enterprise**
 - Centralized log monitoring platform
 - Supports alerts and dashboards
 - Used for event investigation
- **Sysmon**
 - Monitors Windows activities
 - Captures process and network events
 - Improves endpoint visibility
- **Log Sources**
 - Windows Security Logs
 - Sysmon Logs
 - Windows Event Viewer
 - Splunk Universal Forwarder



Architecture



• Components

- Kali Linux – Attack Simulation
- Windows 10 – Monitored Host
- Sysmon – Event Collection
- Splunk UF – Log Forwarding
- Splunk Enterprise – Monitoring & Alerts
- Gmail – Alert Notifications

Authentication Alert

- **Activity Performed**

- Simulated a failed login attempt from Kali Linux.
- Attempted authentication against the Windows system.
- Windows generated Event ID 4625.

- **Detection**

- Splunk detected the failed login activity.
- Alert was triggered automatically.
- Email notification was successfully received.

- **Importance**

- Helps identify unauthorized login attempts.
- Useful for detecting password guessing activity.
- Provides visibility into authentication failures.

```
Home X kali X Windows 10 X
Session Actions Edit View Help
(rev@hackingmachine)-[~]
$ nmap -sV 192.168.228.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-09 11:07 IST
Nmap scan report for 192.168.228.130
Host is up (0.00056s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds?
MAC Address: 00:0C:29:1F:87:6A (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 11.22 seconds

(rev@hackingmachine)-[~]
$ smbclient -L //192.168.228.130 -U fakeuser
Password for [WORKGROUP\fakeuser]:
session setup failed: NT_STATUS_LOGON_FAILURE
```

i	Time	Event
>	6/9/26 11:11:30.647 AM	06/09/2026 11:11:30.647 AM LogName=Security EventCode=4625 EventType=0 ComputerName=DESKTOP-433BODG Show all 61 lines host = DESKTOP-433BODG source = WinEventLog:Security sourcetype = WinEventLog:Security

[SOC ALERT] Failed Login Detected - Failed Login Detection Inbox x

 **kadiyalarevanth28@gmail.com**
to me

11:12 AM (0 minutes ago) ☆ ☺ ↶ ⋮

Failed authentication activity has been detected.

Alert Name: Failed Login Detection
Time: 1780983721

A failed login attempt was detected on the Windows endpoint. This activity may indicate unauthorized access attempts, password guessing, credential misuse, or brute-force activity targeting the system.

Please review the event details in Splunk and investigate the source account, login type, and originating system associated with the authentication failure.

Event Source: Windows Security Log
Event ID: 4625

Alert: [Failed Login Detection](#)

Trigger: Saved Search [Failed Login Detection]: number of events (1)

Trigger Time: 11:12:01 +0530 on June 09, 2026.

[View results](#)

Endpoint Alert

- Activity Performed

- Executed the systeminfo command on Windows.
- Simulated information gathering behaviour.
- Generated Sysmon Event ID 1.

- Detection

- Splunk identified the process execution.
- Alert was triggered based on the detection rule.
- Email notification was generated.

- Importance

- Detects system discovery activities.
- Helps identify reconnaissance behaviour.
- Provides visibility into commands executed on the endpoint.

```
Command Prompt
System Manufacturer: VMware, Inc.
System Model: VMware20.1
System Type: x64-based PC
Processor(s): 2 Processor(s) Installed.
[01]: Intel64 Family 6 Model 183 Stepping 1 GenuineIntel ~2611 Mhz
[02]: Intel64 Family 6 Model 183 Stepping 1 GenuineIntel ~2611 Mhz
BIOS Version: VMware, Inc. VMW201.00V.24866131.B64.2507211911, 21-07-2025
Windows Directory: C:\Windows
System Directory: C:\Windows\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-US;English (United States)
Input Locale: 00004099
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 8,191 MB
Available Physical Memory: 5,304 MB
Virtual Memory: Max Size: 10,111 MB
Virtual Memory: Available: 7,398 MB
Virtual Memory: In Use: 2,713 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \DESKTOP-433BODG
Hotfix(s): 12 Hotfix(s) Installed.
[01]: KB5066130
[02]: KB5066135
[03]: KB5011048
[04]: KB5015684
[05]: KB5020683
[06]: KB5033052
[07]: KB5072653
[08]: KB5071959
```

i	Time	Event
>	6/9/26 11:17:01.568 AM	06/09/2026 11:17:01.568 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-433BODG Show all 38 lines host = DESKTOP-433BODG source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

[SOC ALERT] System Discovery Activity Detected - System Information Enumeration Detected

kadiyalarevanth28@gmail.com

to me

11:18 AM (0 minutes ago)

System discovery activity has been detected.

Alert Name: System Information Enumeration Detected
Time: 1780984081

A process executed the systeminfo command to collect host information. This activity may indicate reconnaissance performed by a user or attacker to gather operating system details, hostname information, system architecture, and patch levels.

Please review the event details in Splunk and investigate the user and process involved.

Event Source: Sysmon
Event ID: 1
Process: systeminfo.exe

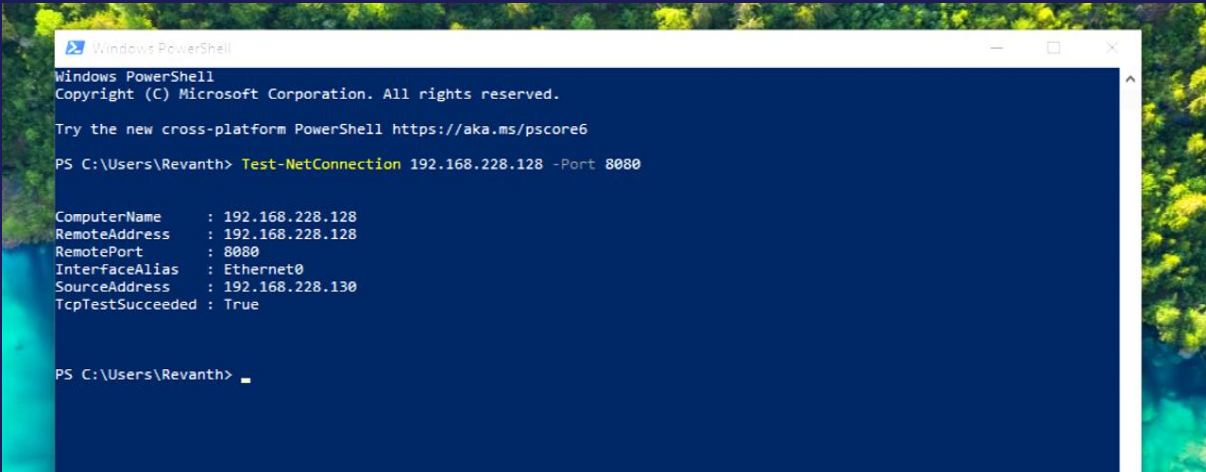
Alert: [System Information Enumeration Detected](#)

Trigger: Saved Search [System Information Enumeration Detected]: number of events (1)

Trigger Time: 11:18:01 +0530 on June 09, 2026.

Network Alert

- **Activity Performed**
 - Used PowerShell to test connectivity to a remote service.
 - Generated a network connection event.
 - Recorded as Sysmon Event ID 3.
- **Detection**
 - Splunk captured the network activity.
 - Alert triggered successfully.
 - Email notification delivered to the analyst.
- **Importance**
 - Detects network reconnaissance activity.
 - Helps identify service discovery attempts.
 - Provides visibility into suspicious network behaviour.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Revanth> Test-NetConnection 192.168.228.128 -Port 8080

ComputerName      : 192.168.228.128
RemoteAddress     : 192.168.228.128
RemotePort        : 8080
InterfaceAlias    : Ethernet0
SourceAddress     : 192.168.228.130
TcpTestSucceeded  : True

PS C:\Users\Revanth>
```

i	Time	Event
>	6/9/26 11:24:19.155 AM	06/09/2026 11:24:19.155 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=3 EventType=4 ComputerName=DESKTOP-433BODG Show all 33 lines host = DESKTOP-433BODG source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

[SOC ALERT] PowerShell Network Reconnaissance Detected

kadiyalarevanth28@gmail.com

to me

11:25 AM (0 minutes ago)

PowerShell-based network reconnaissance activity has been detected.

Alert Name: PowerShell Network Reconnaissance Detected
Time: 1780984501

A PowerShell process initiated a network connection to a remote host. This activity may indicate network reconnaissance, service discovery, lateral movement preparation, or validation of communication channels by an attacker.

Please review the event details in Splunk and investigate the source and destination systems involved.

Event Source: Sysmon Event ID 3
Process: PowerShell.exe

Alert: [PowerShell Network Reconnaissance Detected](#)

Trigger: Saved Search [PowerShell Network Reconnaissance Detected]: number of events (1)

Trigger Time: 11:25:01 +0530 on June 09, 2026.

Log Correlation & Investigation Workflow

- **Event Correlation**

- Combined authentication, endpoint and network logs.
- Linked related events using timestamps.
- Reconstructed the attacker activity timeline.
- Improved visibility during investigation.

- **Analyst Findings**

- Failed login attempt was detected.
- System discovery activity was observed.
- Network reconnaissance activity was identified.
- Events were correlated for further analysis.

SOC Log Correlation Investigation [Save] [Save As] [View] [Create Table View] [Close]

```
(EventCode=4625)
OR (EventCode=1 Image="systeminfo.exe")
OR (EventCode=3 Image="powershell.exe")
| eval Category=case(
  EventCode=4625,"Authentication",
  EventCode=1,"Endpoint",
  EventCode=3,"Network"
)
| table _time Category EventCode Image User ComputerName
| sort _time
```

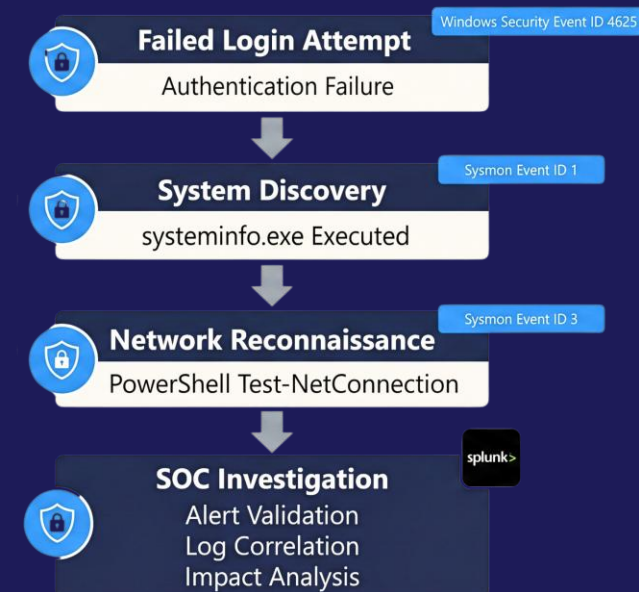
Time range: Last 29 minutes [Search] [Filter]

✓ 3 events (6/9/26 10:59:19.000 AM to 6/9/26 11:28:19.000 AM) [No Event Sampling] [Job] [Pause] [Refresh] [Download] [Smart Mode]

Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page [Format] [Preview: On]

_time	Category	EventCode	Image	User	ComputerName
2026-06-09 11:11:30.647	Authentication	4625			DESKTOP-43800G
2026-06-09 11:17:01.568	Endpoint	1	C:\Windows\System32\systeminfo.exe	NOT_TRANSLATED DESKTOP-43800G\Revanth	DESKTOP-43800G
2026-06-09 11:24:19.155	Network	3	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	NOT_TRANSLATED DESKTOP-43800G\Revanth	DESKTOP-43800G



SOC Dashboard

- **Dashboard Overview**

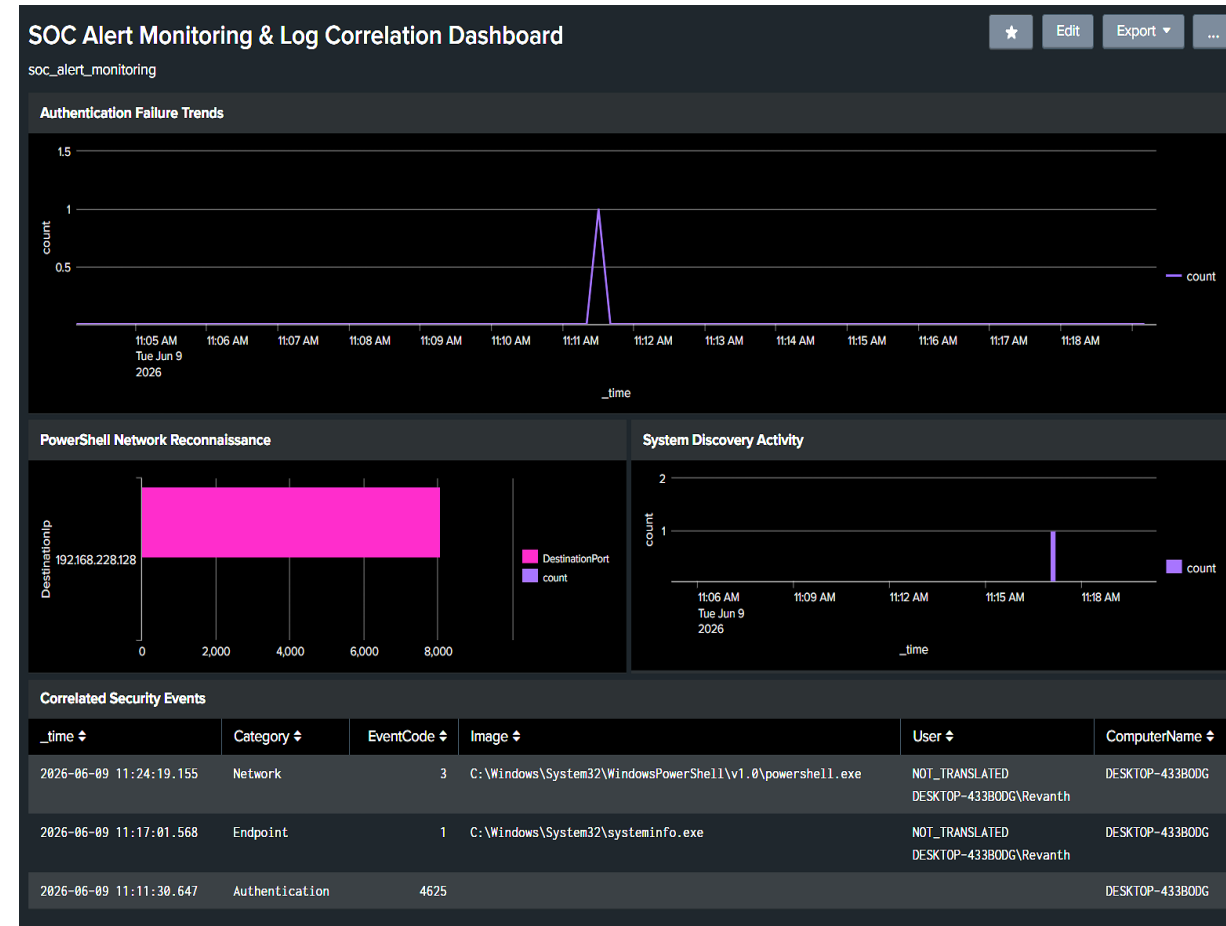
- Created a centralized dashboard for security monitoring.
- Combined authentication, endpoint and network events.
- Provided a single view of security alerts.
- Helped simplify investigation and monitoring.

- **Dashboard Panels**

- Authentication Failure Trends
- System Discovery Activity
- Network Reconnaissance Activity
- Correlated Security Events

- **Benefits**

- Quick visibility into security events.
- Easier alert investigation.
- Faster identification of suspicious activities.



Future Enhancement: AI-Powered SOC Incident Response

- **Objective**

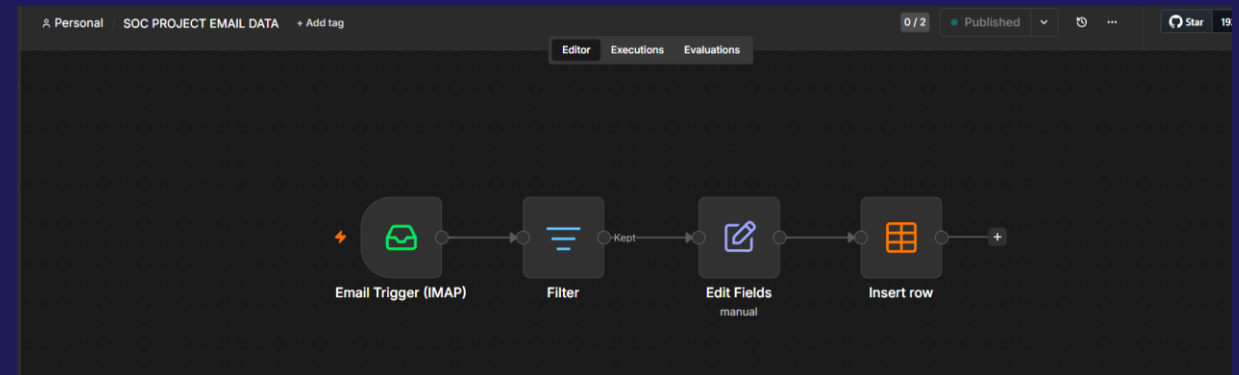
- Automate SOC alert analysis.
- Generate incident reports automatically.

- **Workflow**

- Splunk alerts collected through email.
- Alerts correlated using n8n.
- Google Gemini analyzes related events.
- AI generates an incident report.

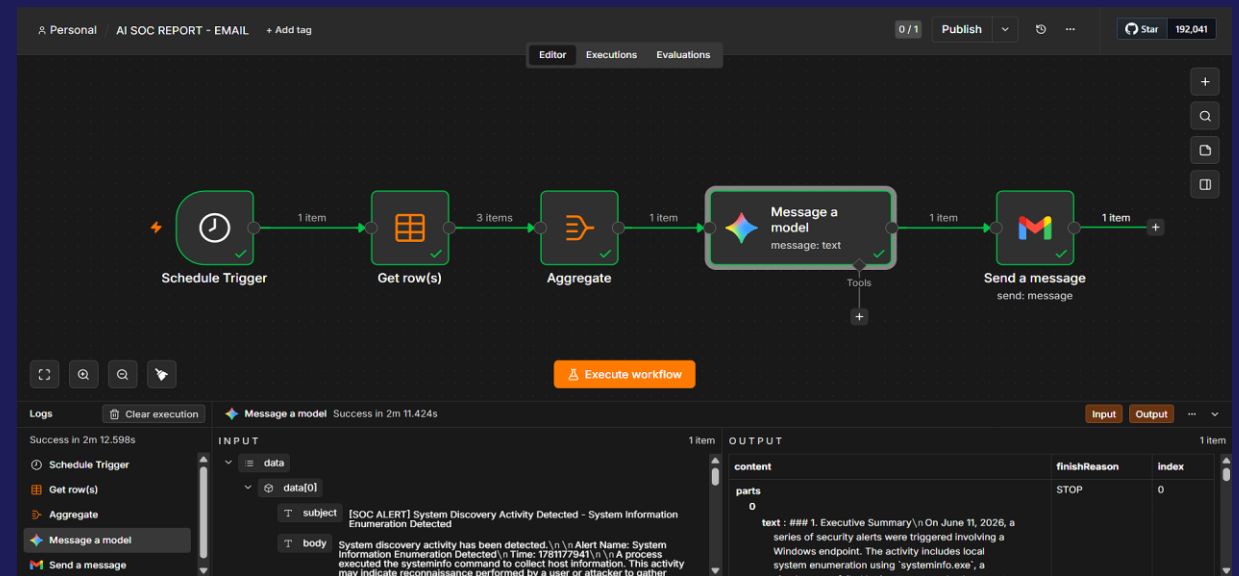
- **Benefits**

- Faster alert triage.
- Reduced manual effort.
- Consistent reporting.



The screenshot shows the n8n interface with a table of data for "soc_alerts". The table has columns for id, subject, body, time, createdAt, and updatedAt. There are three rows of data.

id	subject	body	time	createdAt	updatedAt
1	[SOC ALERT] System Discovery Activity	System discovery activity has been det	Thu, 11 Jun 2026 17:09:02 +0530	2026-06-11T07:39:31.771-04:00	2026-06-11T07:39:31.771-04:00
2	[SOC ALERT] Failed Login Detected - F	Failed authentication activity has been i	Thu, 11 Jun 2026 17:09:02 +0530	2026-06-11T07:39:31.772-04:00	2026-06-11T07:39:31.772-04:00
3	[SOC ALERT] PowerShell Network Rec	PowerShell-based network reconnaiss	Thu, 11 Jun 2026 17:10:02 +0530	2026-06-11T07:40:35.892-04:00	2026-06-11T07:40:35.892-04:00

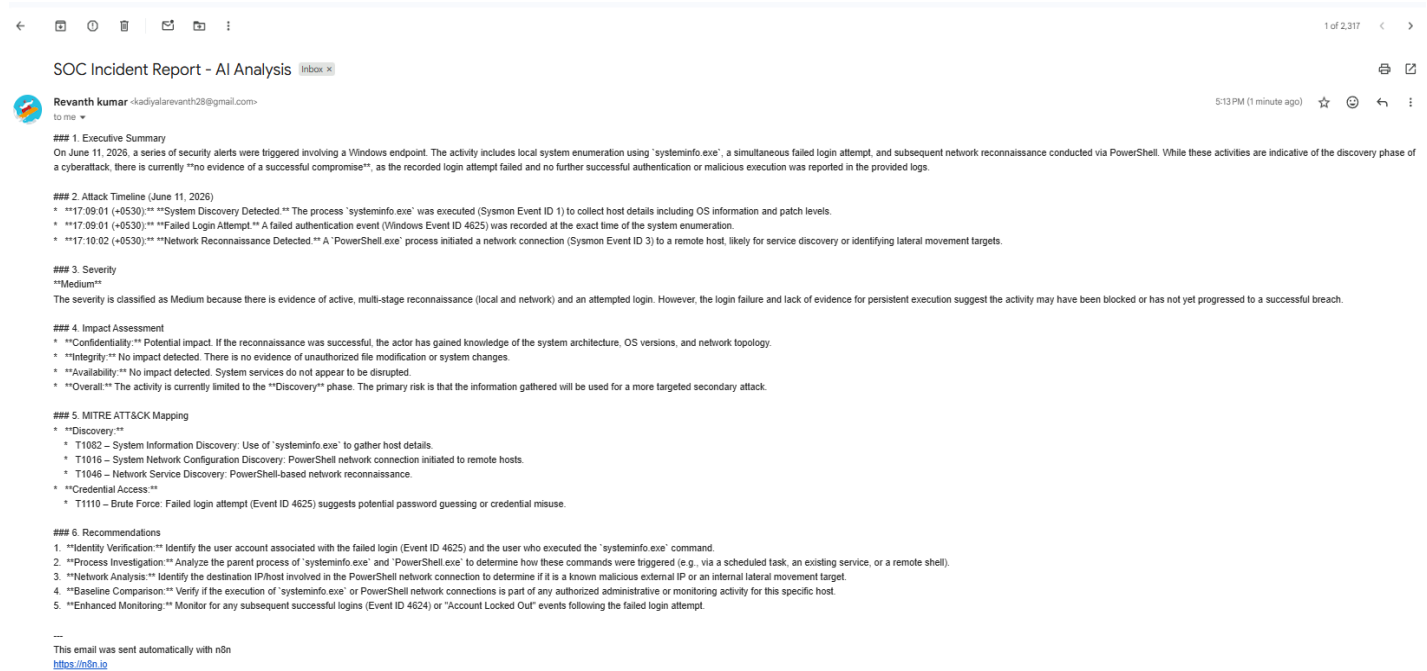


• AI Report Features

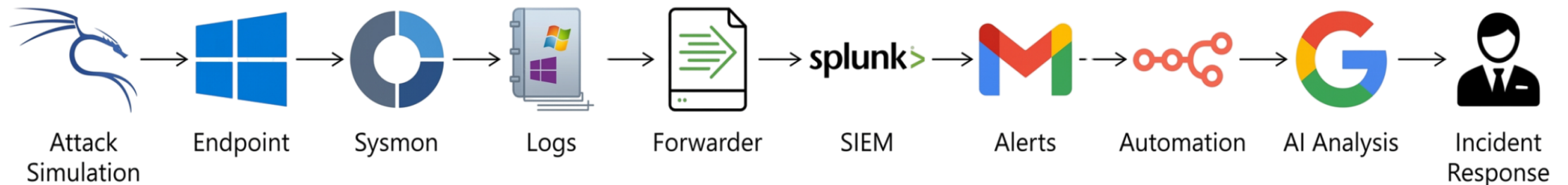
- Executive summary of the incident.
- Attack timeline reconstruction.
- Severity assessment.
- MITRE ATT&CK mapping.
- Investigation recommendations.

• Business Value

- Improves analyst productivity.
- Accelerates incident response.
- Supports faster decision-making.



Final Workflow



Impact Analysis & Recommendations

➤ Impact Analysis

- **Authentication**

- Detected failed login attempts
- Improved account monitoring

- **Endpoint**

- Detected system discovery activity
- Improved endpoint visibility

- **Network**

- Detected network reconnaissance
- Improved activity monitoring

- **Log Correlation**

- Linked multiple events into a single investigation.
- Helped reconstruct attacker behaviour.

➤ Recommendations

- **Alert Tuning**

- Configure thresholds for repeated failed logins.
- Review correlation rules periodically.

- **Reducing False Positives**

- Exclude trusted administrative activities.
- Validate alerts before escalation.

- **Improving Visibility**

- Enable additional Sysmon monitoring.
- Collect logs from more endpoints.

- **Future Enhancement**

- Implement AI-assisted incident reporting.
- Integrate automated ticketing and SOAR workflows.

Conclusion

- Successfully implemented a SOC monitoring environment using Splunk and Sysmon.
- Generated and investigated authentication, endpoint, and network security alerts.
- Correlated multiple events to identify attacker-like behaviour.
- Demonstrated the SOC investigation and triage workflow.
- Enhanced the solution with AI-powered incident reporting.
- **Final Outcome**
 - 3 Security Alerts Generated
 - Security Monitoring Dashboard
 - Log Correlation Report
 - Email Notification System
 - AI-Assisted Incident Reporting

References

- **Documentation**

- [1] Splunk Inc., “Splunk Enterprise Documentation,” Splunk Documentation. [Online]. Available: <https://docs.splunk.com>. [Accessed: Jun. 2026].
- [2] Microsoft Corporation, “Sysmon - System Monitor,” Microsoft Sysinternals Documentation. [Online]. Available: <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>. [Accessed: Jun. 2026].
- [3] Microsoft Corporation, “Security Auditing Overview,” Microsoft Learn Documentation. [Online]. Available: <https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/security-auditing-overview>. [Accessed: Jun. 2026].
- [4] MITRE Corporation, “MITRE ATT&CK Framework,” MITRE ATT&CK Knowledge Base. [Online]. Available: <https://attack.mitre.org>. [Accessed: Jun. 2026].
- [5] IBM Corporation, “What is Security Information and Event Management (SIEM)?,” IBM Security. [Online]. Available: <https://www.ibm.com/topics/siem>. [Accessed: Jun. 2026].
- [6] IBM Corporation, “What is a Security Operations Center (SOC)?,” IBM Security. [Online]. Available: <https://www.ibm.com/topics/security-operations-center>. [Accessed: Jun. 2026].

- **Video References**

- [7] “Splunk Enterprise Installation and Configuration,” YouTube Video. [Online]. Available: https://youtu.be/_nBd39HvOB8. [Accessed: Jun. 2026].
- [8] “Sysmon Configuration and Splunk Integration,” YouTube Video. [Online]. Available: https://youtu.be/P8n0-RKF6_0. [Accessed: Jun. 2026].
- [9] “SOC Monitoring and Alert Investigation Workflow,” YouTube Video. [Online]. Available: <https://youtu.be/LbR5cqqaFVk>. [Accessed: Jun. 2026].

The background is a dark blue gradient with a complex network of glowing blue lines and dots, resembling a molecular structure or a data network. The lines and dots are more concentrated in the lower half of the image, creating a sense of depth and connectivity.

Thank You!